

Northwestern University  
Dept. of Computer Science

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## RESEARCH INTERESTS

My interests lie in *compilers*, specifically how we can leverage modern programming languages to design general-purpose *intermediate representations* that (1) improve the precision of *static analysis* by preserving high-level information, (2) enable novel *optimizations* on the layout and organization of memory, and (3) open the door to *compiler-runtime codesigns* that rethink existing hardware abstractions.

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## EDUCATION

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| <b>Northwestern University</b> , Evanston, Illinois, USA               | 2020 – Present  |
| Ph.D. in Computer Science, <i>Advised by Simone Campanoni</i>          | (Expected 2026) |
| M.Sc. in Computer Science, <i>Advised by Simone Campanoni</i>          | 2023            |
| <b>Rose-Hulman Institute of Technology</b> , Terre Haute, Indiana, USA | 2016–2020       |
| B.Sc. in Computer Engineering and Computer Science                     |                 |

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## PUBLICATIONS

**AI Coding Agents Need Better Compiler Remarks**, *CoDAIM 2026*.

Akash Deo, Simone Campanoni, **Tommy McMichen**.

**Automatic Data Enumeration for Fast Data Collections**, *CGO 2026*.

**Tommy McMichen**, Simone Campanoni.



**Saving Energy with Per-Variable Bitwidth Speculation**, *ASPLOS 2025*.

**Tommy McMichen**, David Dlott, Panitan Wongse-ammatt, Nathan Greiner, Hussain Khajanchi, Russ Joseph, Simone Campanoni.



**Representing Data Collections in an SSA Form**, *CGO 2024*.

**Tommy McMichen**, Nathan Greiner, Peter Zhong, Federico Sossai, Atmn Patel, Simone Campanoni.



**Getting a Handle on Unmanaged Memory**, *ASPLOS 2024*.

Nick Wanninger, **Tommy McMichen**, Simone Campanoni, Peter Dinda.



**Program State Element Characterization**, *CGO 2023*.

Enrico Armenio Deiana, Brian Suchy, Michael Wilkins, Brian Homerding, **Tommy McMichen**, Katarzyna Dunajewski, Peter Dinda, Nikos Hardavellas, Simone Campanoni.



**NOELLE Offers Empowering LLVM Extensions**, *CGO 2022*.

Angelo Matni, Enrico Armenio Deiana, Yian Su, Lukas Gross, Souradip Ghosh, Sotiris Apostolakis, Ziyang Xu, Zujun Tan, Ishita Chaturvedi, Brian Homerding, **Tommy McMichen**, David I. August, Simone Campanoni.



**Fine-Grained Acceleration using Runtime Integrated Custom Execution (RICE)**, *CASES 2019*.

Leela Pakanati, **Tommy McMichen**, Zachary Estrada.

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## INVITED TALKS

**“Representing Data Collections for Analysis and Transformation”**

Languages, Systems, and Data Seminar, *University of California, Santa Cruz*.

Computer Architecture Group Meeting, *University of Cambridge*.

Tech Talk, *Rose-Hulman Institute of Technology*.

Student Seminar Series, *Northwestern University*.

Constellation Workshop, *Northwestern University*.

October 2025

March 2024

October 2023

October 2023

July 2023

**“Towards Collection-Oriented Compilation in LLVM”**

LLVM Developers’ Meeting.

October 2025

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## INDUSTRY EXPERIENCE

**Meta**, Menlo Park, California, USA

Summer 2025

*Software Engineering Intern, Programming Languages and Runtimes, Android Native Compiler Team*

• Improved the representation of ClangIR, an open-source C/C++ MLIR compiler.

- Developed analyses and transformations for C++ move semantics in ClangIR.
- Lead ClangIR open-source development efforts through issue creation and code reviews.

**Texas Instruments**, Dallas, Texas, USA Summer 2019, Summer 2020

*Digital Design Engineering Intern, Embedded Processors, Analytics Team*

- Performed integration testing for hardware implementation of cache coherence protocol.
- Developed coverage metrics for cache coherence testing.
- Implemented automatic generation of RTL and TLM from descriptor files.

**National Instruments**, Austin, Texas, USA Summer 2018

*R&D Software Engineering Intern, Digitizers*

- Designed and implemented FPGA logic for new function generator feature with LabVIEW.
- Added kernel, driver and API support for new function generator feature.
- Implemented full driver stack support for highly-customisable oscilloscope triggers.
- Communicated with multiple teams to add new .NET API entry points.

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## TEACHING EXPERIENCE

**Teaching Assistant**, *COMP\_SCI 322 Compiler Construction*, Prof. Simone Campanoni. Winter 2022

**Resident Tutor**, *Computer Science and Computer Engineering Departments*. Aug. 2019 – May 2020

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## SERVICE

**Student Volunteer**, Workshop on Advancing Theory, Research, and Practice for Generative AI in University-Level Computing Education. 2026

**Artifact Evaluation Committee**, International Conference on Compiler Construction (CC). 2026

**Board Member**, Computer Science PhD Advisory Council, Northwestern University. 2025 – 2026

**Board Member**, Computer Science Social Initiative, Northwestern University. 2021 – 2026

**Member**, CS Ph.D. Orientation Planning Committee, Northwestern University. 2022 – 2025

**Member**, CS Ph.D. Visit Day Planning Committee, Northwestern University. 2022 – 2026

**Student Volunteer**, International Symposium on Microarchitecture (MICRO). October 2022

**Chairperson**, IEEE, Rose-Hulman Institute of Technology student branch. August 2019 – May 2020

**Corresponding Secretary**, Eta Kappa Nu (HKN), Epsilon Eta Chapter. August 2019 – May 2020

**Member**, Eta Kappa Nu (HKN), Epsilon Eta Chapter. May 2018 – May 2020

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## AWARDS AND FUNDING

PhD Student Leadership Award, *Northwestern University – CS Department*. 2026

PhD Student Research Award, *Northwestern University – CS Department*. 2026

LLVM Foundation Student Travel Grant, *LLVM Developers' Meeting*. 2025

NSF Student Travel Grant, *ASPLOS*. 2025

NSF Student Travel Grant, *HPCA/PPoPP/CGO*. 2024

NSF Student Travel Grant, *HPCA/PPoPP/CGO*. 2023

IP/ROP Student Travel Award. 2019

NSF Student Travel Grant, *ESweek*. 2019

IP/ROP Student Project Grant. 2018

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## RESEARCH ADVISING

Leyla Latifova, B.Sc., *Fast Translation Validation for Transformation Pipelines*. 2026 – Present

Akash Deo, M.Sc., *Designing compiler tools for AI-assisted vectorization*. 2025 – Present

Benjamin Ye, M.Sc., *Characterizing differences between LLVM front-ends*. 2025 – Present

Benjamin Ye, B.Sc., *Automatically generating MEMOIR from Rust*. 2024 – 2025